

JEE Main - 8 | JEE 2022 (Gen 1 & 2)

Date: 21/03/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

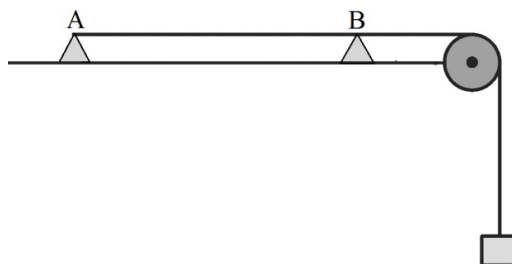
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

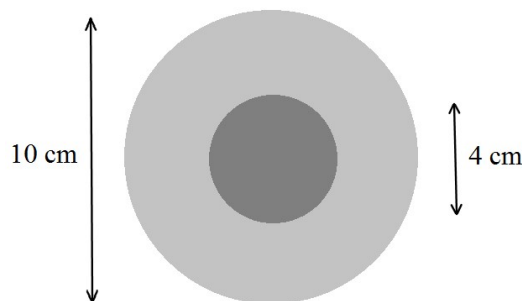
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A particle is projected horizontally from the top of a tower of height H with kinetic energy K . The particle lands on the ground with kinetic energy $2K$. During its flight, the kinetic energy of the particle as a function of its height h above the ground is given by:
 (A) $\left(1 + \frac{h}{H}\right)K$ (B) $\left(2 - \frac{h}{H}\right)K$ (C) $\left(1 + \left(\frac{h}{H}\right)^2\right)K$ (D) $\left(2 - \left(\frac{h}{H}\right)^2\right)K$
2. The radius of gyration of a solid sphere of radius R about an axis P is R . The distance of this axis P from the centre of the sphere is close to:
 (A) $0.60 R$ (B) $0.67 R$ (C) $0.78 R$ (D) $0.83 R$
3. Let R denote the radius of the earth. An object is projected vertically from the surface of the earth with velocity v . Its velocity when it reaches a height R above the surface is $\frac{v}{4}$. The maximum height above the surface that the object reaches is:
 (A) $\frac{19R}{15}$ (B) $\frac{22R}{15}$ (C) $\frac{8R}{7}$ (D) $\frac{10R}{7}$
4. A thin wire of mass per unit length 0.03 kg/m is rigidly clamped at points A and B and a block of weight 75 N is suspended from it as shown. The minimum frequency of a tuning fork with which the segment AB of the wire can resonate is 150 Hz . The length AB (in cm) is:



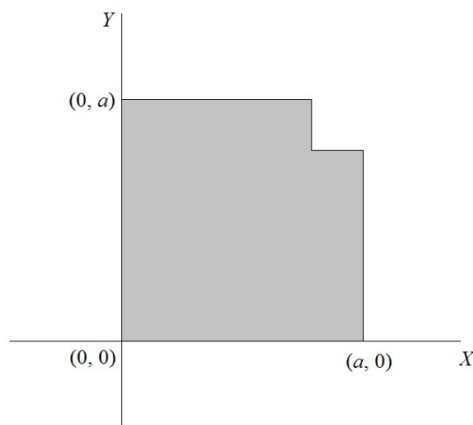
- (A) 16.7 (B) 25.0 (C) 33.3 (D) 50.0
5. A sphere is composed of an inner “core” of diameter 4 cm made of Material A (density 3.0 g/cm^3) and an outer “cladding” of outer diameter 10 cm and inner diameter 4 cm made of Material B (density 0.5 g/cm^3). If the sphere is released into a vessel containing water (density 1.0 g/cm^3):



- (A) It floats with 83% of its volume submerged inside the water
 (B) It floats with 76% of its volume submerged inside the water

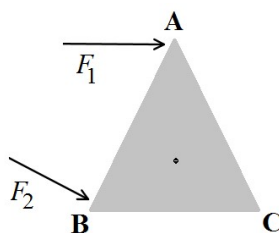
- (C) It floats with 66% of its volume submerged inside the water
(D) It sinks
6. A spherical metal ball of radius r and density ρ is dropped into a tall column of an incompressible liquid of density ρ_L (such that $\rho_L < \rho$). The terminal velocity of the ball is proportional to:
(A) $(\rho - \rho_L) r^2$ (B) $\left(1 - \frac{\rho_L}{\rho}\right) r^2$ (C) $(\rho - \rho_L) r$ (D) $\left(1 - \frac{\rho_L}{\rho}\right) r$
7. When two forces F_1 and F_2 are applied in the same direction, their resultant is 30 N. If the forces are applied at an angle 90° with each other, their resultant is $5\sqrt{26}$ N. If the forces are applied at an angle 60° with each other, their resultant (in Newton) is:
(A) $10\sqrt{7}$ (B) $5\sqrt{31}$ (C) $5\sqrt{33}$ (D) $10\sqrt{5}$
8. A planar equilateral triangle is constructed by joining three identical thin uniform rods. Consider two axes 1 and 2, perpendicular to the plane of the triangle, and passing respectively through the mid-point of a side and the centroid of the triangle. The moment of inertia of the triangle about these two axes, I_1 and I_2 , is in the ratio:
(A) 2:1 (B) 3:2 (C) 5:3 (D) 4:3
9. The position of a particle executing SHM on the X-axis is given by: $x(t) = A \sin^2\left(\omega t - \frac{\pi}{6}\right)$. The mean position of the particle is at:
(A) $x = A$ (B) $x = 0$ (C) $x = \frac{A}{4}$ (D) $x = \frac{A}{2}$
10. It takes 100 kJ heat to raise the temperature of a sample of an ideal monoatomic gas from 20°C to 120°C at constant pressure. The amount of heat needed (in kJ) to raise the temperature of the same sample of gas from 120°C to 220°C at constant volume is:
(A) 60 (B) 166.7 (C) 140 (D) 71.4
11. The fundamental frequency of an organ pipe open at both ends is f_0 . If the length of the pipe is reduced to $\frac{2}{3}$ times and it is closed at one end, its fundamental frequency becomes:
(A) $\frac{3}{2}f_0$ (B) $\frac{3}{4}f_0$ (C) $\frac{1}{3}f_0$ (D) $\frac{2}{3}f_0$
12. A particle initially at rest at the origin starts moving at $t = 0$ with acceleration $a(t) = Pt \hat{i} + Q \hat{j}$, where P and Q are positive constants and t is time. The particle passes through the point (1,2) if:
(A) $P^3 = \frac{4}{9}Q^2$ (B) $P^3 = \frac{16}{9}Q^2$ (C) $P^2 = \frac{9}{4}Q^3$ (D) $P^2 = \frac{9}{16}Q^3$
13. The effective value of acceleration due to gravity on the surface of earth at equator (g') is less than at poles due to the rotation of earth. Now, assume that the time period of rotation of earth about its axis decreases from 24 hrs to 20 hrs. Then:
(A) the value of g' increases (B) the value of g' decreases
(C) the value of g' remains same (D) the value of g' may increase or decrease

14. A smaller square of side length b is cut out from a square lamina of side length a . The resulting body is placed in X-Y plane as shown with two of its sides along the coordinate axes. The centre of mass of this resulting body has the coordinates:

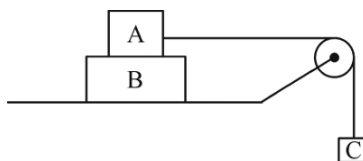


- (A) $\left(\frac{a^2 + b^2 - ab}{2(a+b)}, \frac{a^2 - b^2 + ab}{2(a+b)} \right)$ (B) $\left(\frac{a^2 - b^2 + ab}{2(a+b)}, \frac{a^2 - b^2 + ab}{2(a+b)} \right)$
 (C) $\left(\frac{a^2 + 2b^2 - 2ab}{2(a+b)}, \frac{a^2 + 2b^2 - 2ab}{2(a+b)} \right)$ (D) $\left(\frac{a^2 - 2b^2 + 2ab}{2(a+b)}, \frac{a^2 - 2b^2 + 2ab}{2(a+b)} \right)$

15. An equilateral triangle ABC is cut out of thin metallic sheet. The triangle is placed on a smooth horizontal table and its centroid is fixed. Two horizontal forces F_1 and F_2 are now applied on the triangle at the vertices A and B as shown. The forces are respectively directed parallel to side BC, and perpendicular to side AB. If the triangle remains in equilibrium, the correct relation between F_1 and F_2 is:

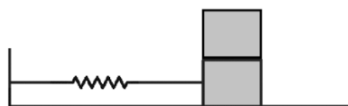


- (A) $2F_1 = \sqrt{3}F_2$ (B) $F_1 = \sqrt{3}F_2$ (C) $\sqrt{3}F_1 = 2F_2$ (D) $\sqrt{3}F_1 = F_2$
16. The blocks A, B and C shown in the figure have mass 3 kg, 1 kg and 2 kg respectively. The pulley and the string are massless. The coefficient of friction between the table and block B is 0.1. The system is released from rest. The minimum value of the coefficient of friction between the blocks A and B such that they do not slip relative to each other is:



- (A) $\frac{1}{6}$ (B) $\frac{2}{9}$ (C) $\frac{5}{12}$ (D) $\frac{7}{18}$

17. When a capillary tube of small diameter is dipped vertically into a liquid such that the length of capillary above the liquid surface is $H = 10$ cm, the liquid rises to a height 4 cm inside the capillary. Now, the capillary is slowly moved down vertically, and hence its length above the liquid surface, H , decreases. Which of these options is correct?
- (A) The liquid leaks from the upper end of the capillary as soon as H decreases below 4 cm
 (B) The liquid doesn't leak even when H decreases below 4 cm and the radius of curvature of meniscus decreases
 (C) The liquid doesn't leak even when H decreases below 4 cm and the radius of curvature of meniscus increases
 (D) The liquid doesn't leak even when H decreases below 4 cm and the radius of curvature of meniscus remains same
18. A particle initially at rest moves in a straight line with constant acceleration. The instantaneous power being delivered to the particle is proportional to s^N , where s is the distance travelled by the particle. Then, N is equal to:
- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 3
19. A monoatomic gas at initial pressure P_0 and initial temperature T_0 is allowed to expand isothermally until its volume doubles. Now, it is compressed adiabatically back to the initial volume. The final pressure and temperature are:
- (A) $2^{2/3} P_0, 2^{2/3} T_0$ (B) $2^{2/3} P_0, 2^{5/3} T_0$ (C) $2^{5/3} P_0, 2^{2/3} T_0$ (D) $2^{5/3} P_0, 2^{5/3} T_0$
20. Two identical blocks, each of mass m , are placed as shown on a frictionless horizontal surface. The lower block is attached to an ideal spring of spring constant k . The coefficient of friction between the blocks is μ . The two blocks are pulled slowly to the right (by the application of a horizontal force on the lower block) such that the spring is stretched by a length L , and the blocks are released. The maximum value of L , such that during their subsequent motion there is no slipping between the blocks, is:



- (A) $\frac{2\mu mg}{k}$ (B) $\frac{\mu mg}{k}$ (C) $\frac{\mu mg}{2k}$ (D) $\frac{\mu mg}{4k}$

SPACE FOR ROUGH WORK

SECTION-2

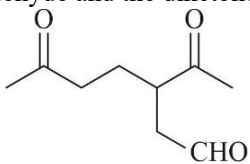
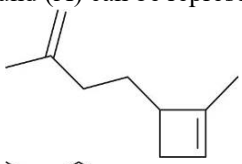
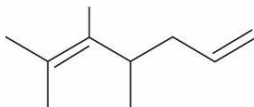
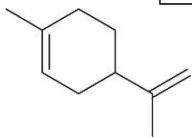
This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. A 10 g ice cube at 0°C is added to an insulated vessel containing 150 g water at 80°C . Assuming that the heat capacity of the vessel is negligible, the final temperature (in $^{\circ}\text{C}$) of the mixture is _____.
(Latent heat of fusion of ice = 336 J/g , specific heat of water = 4.2 J/g)
22. When a longitudinal force is applied on a uniform rod of diameter d made of Material P, the strain produced is 0.01%. If the same magnitude of longitudinal force is applied on a uniform rod of diameter $2d$ of Material Q, the strain produced is 0.02%. The ratio of the Young's modulus of the materials, $\frac{Y_P}{Y_Q}$ is equal to _____.
23. Two cyclists, A and B, are moving on a straight road in the same direction, both at constant velocity 15 km/h with cyclist B 100 m behind cyclist A. A man is walking on the same road in the same direction as the cyclists at constant velocity 5 km/h. At $t = 0$, A overtakes the man. Then, B overtakes the man at $t =$ _____ seconds.
24. A heat engine operating on the Carnot cycle has an efficiency of 25.0%. If the heat absorbed by the engine per cycle is 20 kJ, the heat rejected (in kJ) per cycle is _____.
25. A source emitting sound of a single frequency and an observer are both initially at rest. If the source and the observer start moving directly towards each other at constant velocities which are respectively, 2% and 3% of the speed of sound in air, the percentage change (to the nearest integer) in the observed frequency is _____.

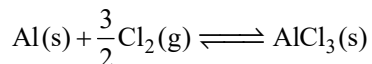
SPACE FOR ROUGH WORK

PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

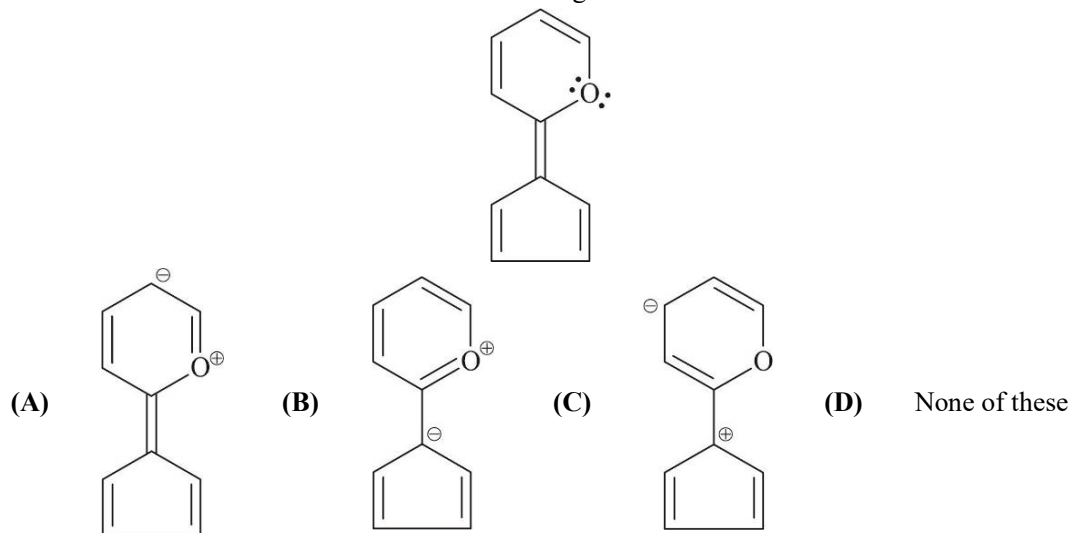
1. **List-1 (Ions)** **List-2 (Shapes)**
- | | | | |
|-----|------------------|-----|----------------------|
| (P) | XeF_5^+ | (1) | Tetrahedral |
| (Q) | SiF_5^- | (2) | Square planar |
| (R) | AsF_4^+ | (3) | Trigonal bipyramidal |
| (S) | ICl_4^- | (4) | Square pyramidal |
| | | (5) | Octahedral |
- | | P | Q | R | S | | P | Q | R | S |
|-----|---|---|---|---|-----|---|---|---|---|
| (A) | 1 | 2 | 5 | 3 | (B) | 1 | 3 | 5 | 2 |
| (C) | 4 | 2 | 1 | 3 | (D) | 4 | 3 | 1 | 2 |
2. The radial probability curve obtained for an orbital wave function (Ψ) has 3 peaks and 2 radial nodes. The valence electron of which one of the following metals does this wave function (Ψ) correspond to ?
 (A) Ca (B) Li (C) K (D) Na
3. The relative rates of effusion of O_2 to CH_4 through a container containing O_2 and CH_4 in 3 : 2 mass ratio will be :
 (A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{3}{4\sqrt{2}}$ (C) $\frac{3}{2\sqrt{2}}$ (D) None of these
4. Compound (A) on hydrogenation consumed 2 equivalents of hydrogen and produces a saturated hydrocarbon. When A is treated with ozone, followed by reduction work up two products are isolated, formaldehyde and the diketone, structure of which is given below :
- 
- Compound (A) can be represented as :
- (A)  (B) 
- (C)  (D) All of these
5. Given below are two statements.
Statement-I : The value of "Biochemical Oxygen Demand" is important for survival of aquatic life.
Statement-II : The optimum value of "Biochemical Oxygen Demand" is 6.5 ppm.
 In light of above statements, choose the most appropriate answer.
 (A) Statement-I is false but Statement-II is true
 (B) Statement-I is true but Statement-II is false
 (C) Both Statement-I and II are false
 (D) Both Statement-I and II are true

6. Cl_2 was introduced in a container made up of aluminium of 5L capacity at 5 atm pressure at 300K. When temperature was raised to 600K then a reaction will start between the $\text{Cl}_2(\text{g})$ and $\text{Al}(\text{s})$ as follows:



If after some time the reaction stops and mass of $\text{AlCl}_3(\text{s})$ formed is 13.35 grams then K_p of the above reaction assuming no change in volume is : [Al = 27 g/mole, Cl = 35.5 g/mole]
[RT = 25 L-atm/mole at 300K]

- (A) $(5.5)^{-3/2}$ (B) $(8.5)^{-3/2}$ (C) $(6.5)^{-3/2}$ (D) $(4.5)^{-3/2}$
7. $\text{B}(\text{OH})_3 + \text{NaOH} \rightleftharpoons \text{Na}[\text{B}(\text{OH})_4]$
The reaction can be made to proceed in forward direction?
- I. By addition of NaOH II. By addition of ethylene glycol $\begin{bmatrix} \text{CH}_2\text{OH} \\ | \\ \text{CH}_2\text{OH} \end{bmatrix}$
- (A) Only I (B) Only II (C) Both I and II (D) None of these
8. A water sample has ppm level concentration of the following anions, $\text{F}^- = 10$, $\text{SO}_4^{2-} = 200$, $\text{NO}_3^- = 100$. The anion/anions that makes the water unsuitable for drinking is/are :
- (A) Only F^- (B) Only SO_4^{2-} (C) F^- and NO_3^- (D) F^- and SO_4^{2-}
9. The most stable canonical structure of the following molecule is :



10. Among LiCl , RbCl , BeCl_2 , MgCl_2 the compounds with greatest and least ionic character respectively are:
- (A) LiCl , RbCl (B) RbCl , BeCl_2 (C) RbCl , MgCl_2 (D) MgCl_2 , BeCl_2
11. At 25°C , the value of pK_b for NH_3 is 4.7. What is the pH of 0.1 M aqueous solution of NH_4Cl with 0.01 M NH_3 ?
- (A) 8.3 (B) 9 (C) 9.5 (D) 10

18. 27 kg of SO_2Cl_2 is reacted with excess of NaOH completely. If the difference between masses of Na_2SO_4 and NaCl produced is x kg, then x is :
 [Given : $M_{\text{SO}_2\text{Cl}_2}^\circ = 135 \text{ g mol}^{-1}$, $M_{\text{Na}_2\text{SO}_4}^\circ = 142 \text{ g mol}^{-1}$, $M_{\text{NaCl}}^\circ = 58.5 \text{ g mol}^{-1}$]
 (Reaction involved $\text{SO}_2\text{Cl}_2 + \text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{NaCl} + \text{H}_2\text{O}$)
 (A) 5 kg (B) 3 kg (C) 2 kg (D) None of these
19. Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar?
 (A) $\text{Ca} < \text{Ba} < \text{S} < \text{Se} < \text{Ar}$ (B) $\text{S} < \text{Se} < \text{Ca} < \text{Ba} < \text{Ar}$
 (C) $\text{Ba} < \text{Ca} < \text{Se} < \text{S} < \text{Ar}$ (D) $\text{Ca} < \text{S} < \text{Ba} < \text{Se} < \text{Ar}$
20. Which of the following is a disproportionation reaction?
 (A) $\text{Cl}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{ClO}^-(\text{aq}) + \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\ell)$
 (B) $\text{Cl}_2(\text{g}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{I}_2(\text{s})$
 (C) $2\text{Fe}(\text{s}) + 3\text{H}_2\text{O}(\ell) \xrightarrow{\Delta} \text{Fe}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$
 (D) $2\text{H}_2\text{O}(\ell) + 2\text{F}_2(\text{g}) \rightarrow 4\text{HF}(\text{aq}) + \text{O}_2(\text{g})$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

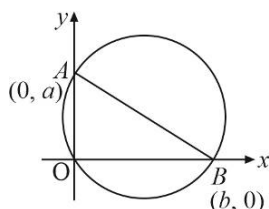
21. How many of the following metal can be identified by flame test ?
 (a) Li (b) Na (c) K
 (d) Ca (e) Ba
22. Find energy level of Li^{2+} having velocity of the electron equal to $2.18 \times 10^6 \text{ m/sec}$.
23. Most probable speed of a gas at 150 K is equal to the root mean square speed of O_2 gas at 400 K. Molecular mass of the gas in g/mol is:
24. Total number of isomers (including stereoisomers) obtained by monochlorination of 2, 2, 4-trimethylpentane is/are.
25. If value of the Avogadro number is $6.023 \times 10^{23} \text{ mol}^{-1}$ and the value of Boltzmann constant is $1.380 \times 10^{-23} \text{ JK}^{-1}$, then the number of significant digits in the calculated value of the universal gas constant is:

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- From the point $P(16, 7)$ tangents PQ and PR are drawn to the circle $x^2 + y^2 - 2x - 4y - 20 = 0$. If C be the center of the circle, then area of quadrilateral PQCR is:
 (A) 450 sq units (B) 15 sq units (C) 50 sq units (D) 75 sq units
- The equation of the lines through the point $(3, 2)$ which makes an angle of 45° with the line $x - 2y = 3$, are:
 (A) $3x - y = 7$ and $x + 3y = 9$ (B) $x - 3y = -7$ and $3x + y = 11$
 (C) $2x + y = 8$ and $x - 2y = -1$ (D) $x + 2y = 7$ and $2x - y = 4$
- In the expansion of $(2 - 3x^3)^{20}$, if the ratio of 10th term 11th term is $\frac{45}{22}$, then x is equal to:
 (A) $-\frac{2}{3}$ (B) $-\frac{3}{2}$ (C) $-\sqrt[3]{\frac{2}{3}}$ (D) $-\sqrt[3]{\frac{3}{2}}$
- Locus of the feet of the perpendicular drawn from either foci on a variable tangent to the hyperbola $16y^2 - 9x^2 = 1$ is:
 (A) $x^2 + y^2 = 9$ (B) $x^2 + y^2 = 1/9$ (C) $x^2 + y^2 = 7/144$ (D) $x^2 + y^2 = 1/16$
- Five digit numbers with distinct digits are formed by using the digits, 5, 4, 3, 2, 1, 0. The number of those numbers which are multiples of 3, is:
 (A) 720 (B) 240 (C) 216 (D) 120
- The foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. Then the equation of the hyperbola with eccentricity $\sqrt{2}$ is:
 (A) $x^2 - y^2 = 12$ (B) $x^2 - y^2 = 8$ (C) $x^2 - y^2 = 4$ (D) $y^2 - x^2 = 12$
- The normal chord of a parabola $y^2 = 16x$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is:
 (A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$
- Let a and b represent the length of a right triangle's legs. If D is the diameter of a circle circumscribed on the triangle, then D equals:



- (A) $a + b$ (B) $2(a + b)$ (C) $\frac{1}{2}(a + b)$ (D) $\sqrt{a^2 + b^2}$

9. A straight line is drawn through P(4, 5) to meet the axis of x and y at A and B respectively. If the rectangle OACB is completed, then locus of C, is:
 (A) $\frac{x}{4} + \frac{y}{5} = 1$ (B) $\frac{5}{x} + \frac{4}{y} = 1$ (C) $\frac{4}{x} + \frac{5}{y} = 1$ (D) $\frac{x}{5} + \frac{y}{4} = 1$
10. The vertices of a triangle are $(\sqrt{3}, 1), (-2 \cos \theta, 2 \sin \theta)$ and $(2 \sin \theta, 2 \cos \theta)$, where $\theta \in R$. The locus of orthocenter of the triangle is:
 (A) $(x - \sqrt{3})^2 + (y - 1)^2 = 4$ (B) $(x - \sqrt{3})^2 + (y - 2)^2 = 4$
 (C) $(x - \sqrt{3})^2 + (y - 1)^2 = 8$ (D) $(x - \sqrt{3})^2 + (y - 2)^2 = 8$
11. If $f(x) = \sum_{k=3}^{\infty} \sin^k(x)$, then the value of $f\left(\frac{\pi}{3}\right)$ is equal to:
 (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{3\sqrt{3}}{8 - 4\sqrt{3}}$
12. If α, β are the roots of equation $x^2 + x - 1 = 0$, then the value of $\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2 + \beta^2 + \alpha + \beta}$ is equal to:
 (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 2 (D) -2
13. Number of ordered pairs (x, y) of real numbers satisfying the system of equations
 $\sin x = \sin 2y$ and $\cos x = \sin y$ where $0 \leq x \leq \frac{\pi}{2}$ and $0 \leq y \leq \frac{\pi}{2}$, is:
 (A) 1 (B) 2 (C) 3 (D) 4
14. If the points of intersection of lines $L_1 : y - m_1x + k = 0$ and $L_2 : y - m_2x + k = 0$ ($m_1 \neq m_2$) lies inside a triangle formed by the lines $2x + 3y = 1, x + 2y = 3$ and $5x - 6y - 1 = 0$, then true set of values of k are:
 (A) $\left(\frac{1}{3}, \frac{3}{2}\right)$ (B) $\left(-\frac{3}{2}, -\frac{1}{3}\right)$ (C) $\left(0, \frac{3}{2}\right)$ (D) $\left(\frac{-3}{2}, 0\right)$
15. If the value of the sum $29\binom{30}{0} + 28\binom{30}{1} + 27\binom{30}{2} + \dots + 1\binom{30}{28} - 1$, where $\binom{n}{r} = {}^nC_r$, is equal to $k2^{32}$, then the value of k is equal to:
 (A) 7 (B) 14 (C) 5/2 (D) 7/2
16. All possible three digits even numbers which can be formed with the condition that if 7 is one of the digit, then 8 is the next digit is:
 (A) 5 (B) 325 (C) 373 (D) 374
17. If $1 + x^2 = \sqrt{3}x$, then $\sum_{n=1}^{24} \left(x^n - \frac{1}{x^n}\right)^2$ is equal to:
 (A) 0 (B) 48 (C) -24 (D) -48
18. The sets $S_1, S_2, S_3, \dots$ are given by $S_1 = \left\{\frac{2}{1}\right\}, S_2 = \left\{\frac{3}{2}, \frac{5}{2}\right\}, S_3 = \left\{\frac{4}{3}, \frac{7}{3}, \frac{10}{3}\right\}, S_4 = \left\{\frac{5}{4}, \frac{9}{4}, \frac{13}{4}, \frac{17}{4}\right\}, \dots$
 Then, the sum of the numbers in the set S_{25} is:
 (A) 320 (B) 322 (C) 324 (D) 326

19. A lady gives a dinner party for six guest. The number of ways in which they may be selected from among ten friends, if two of the friends will not, attends the party together is:
 (A) 112 (B) 140 (C) 164 (D) None of these
20. The constant term in the expansion of $\left(x^2 + \frac{1}{x^2} + y + \frac{1}{y}\right)^6$ is:
 (A) 420 (B) 400 (C) 380 (D) 360

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08)

21. Find the product of values of x satisfying the equation $1 - \log_9(x+1)^2 = \frac{1}{2} \log_{\sqrt{3}}\left(\frac{x+5}{x+3}\right)$.
22. Consider $g_1, g_2, g_3, \dots, g_n$ are in increasing G.P. such that $\frac{1}{g_1} + \frac{1}{g_n} = \frac{33}{64}$, $g_2 g_{n-1} = 128$ and $\sum_{i=1}^n g_i = 126$, then value of n is _____.
23. Tangent and normal are drawn at the upper end (x_1, y_1) of the latus rectum P with $x_1 > 0$ and $y_1 > 0$, of the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$, intersecting the transverse axis at T and G respectively. Find the area of the triangle PTG.
24. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ ($a, b, c, d \in R$) has 4 non real roots, two with sum $4 + 5i$ and the other two with product $12 + i$. Find the value of 'b'.
25. The total number of ways of selecting 6 balls out of 20 identical red balls, 10 identical blue balls and 6 identical green balls is _____.

SPACE FOR ROUGH WORK